NB-IoT RACH Revision — Problem Definition, Literature Review & Research Gaps

Purpose and evidence status

Working revision document — not a claim of new experimental results. This document reassesses the rejected manuscript, *Lightweight Q-Learning Based Joint RACH Optimization for Ultra-Dense NB-IoT Smart Meter Networks*, against the DELCON reviews and the supplied research repository. It provides a revised problem definition, a replacement literature-review narrative, evidence-bounded research gaps, and the validation needed to make those gaps defensible.

Sources and coverage

- **M1:** Uploaded *Conference_Manuscript.pdf*, seven pages. Section and page references below refer to this supplied manuscript.
- **M2:** Uploaded DELCON review PDF, three pages. Reviewer A's substantive comments appear on page 2; Reviewer B's detailed comments continue on page 3.
- **Repository:** Both literature-review DOCX files, the MATLAB simulator, and the three project-description images were inspected. The research folder contains 39 PDFs, including duplicate files and alternate versions. Titles/abstracts were screened to establish relevance; the central papers' methods, relevant passages, and/or conclusions were examined more closely. This is a targeted review of the supplied collection, not a systematic review of all published research.
- Repository revision identified during inspection: `da8ab1705703dd8c4ce8eeb2e8c78bf0d69e5be4`. The source links below use this revision for traceability. The archive was downloaded from `main` during this session.
- One important supplement outside the repository was checked: the publisher abstract and introduction for Alcaraz et al.'s model-based NB-IoT RL paper,

already cited in M1. Its complete paywalled methods were not available in the retrieved page.

- **No MATLAB training, baseline reproduction, or new performance experiment was run.** Code findings are from static inspection. Existing numerical results remain manuscript-reported values, not independently reproduced findings.

1. What the original work is trying to solve

1.1 Original engineering problem

The manuscript considers one NB-IoT cell serving a large population of smart meters. Normal reporting is supplemented by concentrated access demand associated with events such as demand-response acknowledgements and outage restoration. When many requests enter contention together, devices choose overlapping NPRACH resources, repeat access attempts, and accumulate waiting time.

The proposed solution places a tabular Q-learning agent at the **eNB**, not inside each meter. It uses discretised load, collision history, and backlog to choose three controls: an ACB-related probability, a backoff setting, and an NPRACH resource setting. The intended benefit is coordinated adaptation without the model size of a neural-network controller. M1 describes 150 states, 120 joint actions, and a 72,000-byte float32 Q-table, and reports an attempt-related success value of 0.592 and a mean delay estimate of 1,534 ms for 50,000 meters. These are claims to revalidate, not results to carry unchanged into a revised abstract. [M1, pp. 1–6]

1.2 What is useful and should be retained

1. **The application:** event-driven access congestion in smart-meter networks is a meaningful engineering problem; Reviewer A also recognises its practical relevance. [M2, p. 2]
2. **The coordinated-control question:** admission, retry timing, and resource allocation interact. It is reasonable to test their combined effect.
3. **The low-complexity design preference:** tabular control is a useful candidate if its state aggregation does not hide deadline-critical information.

4. **The network-side architecture:** it matches the controls being studied. Meter CPU limitations are not the main justification for an eNB-resident agent.

1.3 What must change in the scientific argument

The revised paper should not begin from “Q-learning is new” or “nobody jointly optimises access parameters.” Its central question should instead be:

> **Under explicitly defined AMI traffic, observability, resource, and signalling constraints, can a compact network-side controller improve the fraction of requests completed before their deadlines, compared with correctly implemented analytical and learning baselines, without hiding deferred or unfinished requests?**

This is a proposed research question. Its novelty and performance remain to be demonstrated.

2. What the reviewers actually require

Review issue	Interpretation	Required revision and evidence
A: Baselines differ between Section V-A, Table II, and Figures 3–4	The reader cannot identify a single experiment being reported	Define one baseline registry with exact method names, source, code entry point, parameters, and run identifiers. Generate every table and figure from those runs.
A: DQN/DDQN/PPO are labelled proxies	Approximations do not establish superiority over the named algorithms	Remove proxy curves from comparative evidence. Implement and train an actual selected DRL baseline, or narrow the comparison and claims.
A: Collision and success columns are complements	Two columns may report the same binary outcome	State the event and denominator. Prefer on-time completion, unresolved backlog, and tail delay as independent outcomes.
A: Metric `e` is undefined	Energy-related claims are not interpretable	Define a physical energy model with units or relabel the quantity as a dimensionless attempt proxy.
A: Novelty is incremental	Merely adding a control dimension to	Establish the closest prior art and test a specific new hypothesis: deadline

Review issue	Interpretation	Required revision and evidence
	tabular RL is insufficient	accounting, deployable action timing, or robustness under partial observation.
B: Compute-reward step is missing from the text and disconnected in Figure 2	The algorithm is not reproducible from the description	Write the exact reward equation, measurement timing, units/normalisation, and next-state construction; connect reward computation to the Q update in a manually verified diagram.
B: Algorithm 1 is referenced but absent	A promised method component is missing	Include complete pseudocode matching the executable simulator.
A/B: Broken reference, placeholder citation, spacing, and cropped Figure 4	Submission quality undermines confidence in the technical work	Rebuild cross-references and vector figures; verify the exported PDF at full size and in two-column layout.
B: Indicator-function notation is called a hallucination	The notation needs clarification, but the mathematical concept itself is valid	Define the indicator explicitly as 1 when the condition holds and 0 otherwise. More importantly, correct the SLA event and denominator; do not discard valid notation merely because the review objects to it.

Important citation correction: Reviewer A suggests replacing LE-URC's `[Rxx]` with `[13]`. In the supplied manuscript, however, reference [13] is Sharma and Wang's collaborative distributed Q-learning paper, not an identified LE-URC source. The proper fix is to verify and cite the actual LE-URC origin, not mechanically substitute a number. Jiang et al.'s repository paper explicitly discusses LE-URC. [M1, p. 7; M2, p. 2][1]

3. Additional issues found in the repository

These are **audit findings**, not additional reviewer quotations. They explain why rewriting the introduction alone would not resolve the rejection.

3.1 Request conservation is not established

The MATLAB queue update is `Q_back = max(0,Q_back-rel) + N_coll`, followed by a cap at `N_UE`. Barred contenders are not explicitly retained in a barred-request queue.

Comments say they return through background arrivals, but a fixed background arrival rate does not track the identity, number, or age of barred requests. The cap can also suppress excess backlog without recording drops. Consequently, the implementation does not substantiate M1's claim that every generated request remains until successful access.[2]

Revision: preserve every request in exactly one explicit status—waiting/barred, retry backoff, in an access procedure, completed, or explicitly dropped. Record terminal outcomes rather than silently clipping counts.

3.2 The recorded denominator is not the realised number of attempts

`rach_step` draws admitted devices stochastically, but the caller reconstructs `M_pass` as `round(M_cont*(1-p_ACB))`. Successes and collisions come from the random realisation; the denominator comes from its rounded expectation. Attempt-level metrics should instead use the actual admitted count returned by the simulator. Also, a high success rate among admitted attempts can coexist with poor service to generated requests.[3][4]

3.3 Delay and SLA are not measured per request

The code constructs delay from analytical ACB, collision/backoff, and queue terms; it does not measure arrival-to-completion latency for each request. Its SLA statistic counts active intervals whose estimated delay exceeds a threshold. That is not the fraction of application requests missing their deadlines. The code also uses `>` in the reward and `>=` in the SLA counter; the boundary convention must be consistent.[5][4]

Revision: keep the analytical delay approximation only as a labelled approximation and cross-check it against timestamped request delays. Zero observed violations would still be an empirical result, not a hard guarantee.

3.4 The implemented reward differs from the manuscript's description

The code uses `N_suc/N_p - Q_back/N_UE - 0.5*double(D > D_SLA)`. This is resource-normalised successful access with backlog and estimated-delay penalties, not simply the number of successful procedures. Maximising `N_suc/N_p` is not identical

to maximising total completed requests when `N_p` is controllable. A soft penalty does not enforce a hard deadline constraint.[6]

3.5 The proxy concern is confirmed by the code

`run_drl_baseline` explicitly states that it is not training DQN/DDQN/PPO; it substitutes parameterised behaviour. `gen_drl_curve` produces synthetic convergence curves. A separate optional fast-plot branch contains locked result values and artificial seed variation; `FAST_PLOT_MODE` is currently false, so this does **not** establish that those values produced the submitted results. It does mean the plotting path must be separated from scientific evaluation. Labels asserting GPU-required inference are also not measurements of implemented networks and should be removed.[7]

3.6 LE-URC is not faithfully demonstrated

The branch named `leurc` changes the barring quantity from backlog while retaining a fixed NPRACH resource argument in the inspected baseline routine. This does not establish reproduction of the LE-URC resource-configuration method discussed by Jiang et al. Rename it as the heuristic actually implemented until a source-faithful implementation is available.[8][1]

3.7 Model, timing, and learning assumptions need reconciliation

- The manuscript adds periodic and burst arrivals; the code uses burst-only arrivals for ten slots, then a constant background rate. At 50,000 devices, ten requests per 50 ms is 200 requests/s. If each of those devices reports uniformly once every 15 minutes, the corresponding mean is about 55.6 requests/s. The present code therefore should not be described as a demonstrated 15-minute per-meter reporting process.
- The code uses `p_ACB` as a **barring** probability: devices pass when a uniform random draw exceeds it. The paper must distinguish this from an admission probability or a standards-defined ACB factor.
- The claimed legal action set, 50 ms resource timing, and per-interval broadcast/update feasibility require an explicit mapping to the chosen NB-IoT release and signalling mechanism. Listing TS 36.211 and TS 36.321 is not sufficient to prove all actions and update rates are deployable.

- The Q update uses a constructed next-state proxy rather than the next observation actually produced by the subsequent traffic/retry step. Retry release is also calculated before action selection and recalculated after BI changes. Both require a consistent event timeline.
- The implemented greedy action search and TD target scan 120 actions: complexity is O(|A|), not a single scalar lookup as the paper implies. A precomputed fixed policy can give O(1) action lookup; online learning still has additional work. The 72,000-byte table is about 70.3 KiB, excludes simulator/controller overhead, and would change if richer state information is added.
- A fixed learning rate of 0.25 does not, by itself, satisfy the usual diminishing-step-size convergence conditions. Final training episodes with residual exploration are not equivalent to an independent, frozen-policy test.

These findings follow from the inspected MATLAB implementation; they should be resolved against the actual code and logs used for submission.[7]

4. Revised problem definition

4.1 Manuscript-ready problem statement

> Advanced Metering Infrastructure produces a mixture of periodic reports and event-triggered messages whose access demand can become concentrated in time. In a dense NB-IoT cell, this concentration couples access admission, retry timing, and NPRACH resource allocation: aggressive admission can increase contention, excessive barring can postpone useful service, and additional access resources consume capacity otherwise available for data transmission. Existing adaptive and learning-based methods already address several of these mechanisms. The unresolved question considered here is therefore not whether reinforcement learning can reduce collisions, but whether a compact, network-side controller can improve deadline-compliant request completion under a specified AMI workload and realistic control constraints. The controller must account for deferred, retried, failed, and unfinished requests; operate from measurements available to the eNB; and be evaluated against reproducible analytical and learning baselines under the same resource budget. Performance is assessed using on-time completion, tail access delay,

> unresolved backlog, and controller cost rather than attempt-level success alone. Where offered traffic exceeds feasible service capacity, the system must report overload and unmet demand rather than claim an unconditional latency guarantee.

4.2 Scope and service endpoint

Recommended initial scope: one NB-IoT cell and one explicitly specified CE/resource configuration, followed by CE-stratified stress tests if those are implemented. Treat 10,000, 50,000, and 100,000 as candidate population stress points, not proof that all such populations can be served within one deadline.

Define the endpoint precisely:

- **Preamble success:** a successful initial contention outcome.
- **Random-access completion:** completion of the modelled access procedure, including the relevant response/contention stages.
- **Application delivery:** successful meter-message delivery to the stated network/application endpoint.

These are not interchangeable. If only singleton preambles are simulated, call the outcome preamble success and do not claim application SLA compliance. Firmware delivery itself is not an uplink-access workload; only its explicitly modelled uplink signalling or acknowledgements belong in this problem.

4.3 Request accounting and delay

Give each generated request an ID, class, creation time, deadline, retry counter, and current status. Multiple messages from a meter must either have distinct request IDs or follow an explicit coalescing rule.

For cumulative arrivals A, completions C, explicit drops X, and pending requests B, enforce the accounting invariant:

$$A_{\mathrm{cum}}(t) = C_{\mathrm{cum}}(t) + X_{\mathrm{cum}}(t) + B(t).$$

Here B includes barred requests, retry timers, and in-progress procedures. It is not merely the collision queue. Backoff moves a request between pending states; it does not remove that request from the system.

For completed request i, measure:

$$D_i = t_i^{\text{complete}} - t_i^{\text{arrival}}.$$

At test horizon H, form a deadline-matured cohort E(H): requests whose deadlines have elapsed by H. Define on-time completion and deadline-miss rates on that same cohort:

$$P_{\text{on}} = \frac{\sum_{i \in E(H)} \mathbf{1}\{t_i^{\text{complete}} \le t_i^{\text{deadline}}\}}{|E(H)|}, \qquad P_{\text{miss}} = 1 - P_{\text{on}}.$$

Count requests still pending after their deadline, dropped without timely completion, or completed late as misses. Report requests whose deadlines have not matured separately; do not silently exclude unfinished requests to make latency appear better. Report median, p95, and p99 delay for completed requests **alongside** completion and miss rates, so survivorship bias remains visible.

4.4 Controller and objective

Use an action vector such as `(admission probability, permitted retry/backoff setting, permitted NPRACH configuration)`. Keep the existing three-control concept, but map every candidate action to an exact supported mechanism, legal value, update interval, and signalling delay. If this mapping fails, narrow the action set or label the model as abstract rather than standards-compliant.

Prefer a finite-horizon objective that maximises on-time completed requests under a fixed time-frequency resource budget. A chance constraint such as `P(D_i > d_class) ≤ ε_class` may be studied only after deadlines, acceptable miss rates, and workload feasibility have been specified. A reward penalty is a mechanism for optimisation—not proof that such a constraint holds.

For persistent traffic, assess stability only when the offered load is within a supportable region. For finite overload bursts, report backlog-drain time and remaining requests. `Q(t) ≥ 0` is a physical condition, not evidence of queue stability.

4.5 Feasibility before learning

A simple diagnostic exposes the importance of this distinction. **Under the code's simplified abstraction** of one opportunity every 50 ms, 48 orthogonal choices per opportunity, one singleton success per choice, and no parallel resources, 30 seconds provides 600 opportunities. Even an impossible collision-free schedule could complete at most `600 × 48 = 28,800` such one-step accesses. Therefore, completing a simultaneous 50,000-request batch within that common 30-second horizon is impossible in that abstraction, before accounting for retransmissions or later protocol stages.

With independent uniform random choice, expected singleton successes for M attempts over N choices are `M(1−1/N)^(M−1)`. Around M=N=48 this is about 17.8 successes per opportunity, or roughly 10.7 thousand over 600 opportunities at continuously well-matched load. This is an illustrative contention-capacity calculation, **not a bound for every real NB-IoT deployment**. Multiple CE resources, carriers, different opportunity timing, and capture change the assumptions. The revised paper must specify those resources before selecting a deadline and target completion fraction.

5. Updated literature review — replacement narrative

5.1 Adaptive barring and resource configuration precede this work

Random access under massive, concentrated arrivals is an established research problem. Duan et al.'s D-ACB work estimates congestion and adapts barring; importantly, it includes both fixed and dynamic preamble-allocation variants. It should not be reduced to a method that universally fixes preamble resources. This analytical line provides essential non-learning baselines and demonstrates that useful adaptation is not unique to RL.[9]

The 2024 survey by Iiyambo, Hancke, and Abu-Mahfouz places congestion control within a broader NB-IoT RA landscape, including conventional contention control, learning-based approaches, and grant-free alternatives. It is useful for organising the review, but individual algorithm claims should be supported by the original papers. The revised paper should therefore separate admission control, retry/backoff, resource configuration, prediction, and alternative access protocols rather than classifying the entire literature as static or single-parameter.[10]

5.2 Tabular RL and multi-parameter NB-IoT optimisation are established

Tello-Oquendo et al.'s LTE-A work uses reinforcement learning to adapt the ACB barring rate for simultaneous H2H/M2M traffic. Jiang et al.'s *Deep Reinforcement Learning for Real-Time Optimization in NB-IoT Networks* evaluates tabular Q-learning, linear approximation, and DQN, then studies action aggregation and cooperative multi-agent DQN for a multi-parameter, multi-group NB-IoT setting. Its nine action variables cover resource/repetition/preamble configuration across three groups. These studies directly rule out portraying tabular RL for RACH, network-side learning, or multi-parameter NB-IoT control as new in themselves. [11][1]

Jiang et al.'s NB-IoT study is especially important because it jointly considers resources for random access and data transmission, and explicitly recognises partial observability. The relevant comparison is consequently whether a smaller policy can retain useful performance while making a transparent trade-off in observability, action granularity, training cost, and request-level QoS. A smaller Q-table alone does not demonstrate a superior solution.[1]

5.3 Backoff and multi-objective learning cannot be claimed absent

Jiang et al.'s *A Decoupled Learning Strategy for Massive Access Optimization in Cellular IoT Networks* explicitly studies ACB, backoff, and distributed queuing, combines success, delay, and energy objectives, and uses prediction with cooperative learning for access-control configuration. Separately, Ziqi Chen's 2020 thesis describes a base-station DRL algorithm controlling ACB and backoff under burst activation and heterogeneous QoS. Thus, the supplied collection itself contradicts the earlier review's blanket assertion that backoff has been universally ignored.[12][13]

Bui and Pham jointly tune the ACB barring factor and mean barring time using dueling DQN to balance delay and energy satisfaction. **ACB barring time is not the same mechanism as the retry Backoff Indicator.** This distinction matters: the study demonstrates prior joint temporal access control, but it should not be incorrectly coded in a comparison table as direct BI optimisation. Hadjadj-Aoul and Ait-Chellouche use TD3 for NB-IoT blocking-factor control under uncertain

load. Together these works motivate accurate control-variable taxonomy and actual algorithm comparisons rather than proxies based on algorithm names.[14] [15]

5.4 Prediction and closed-loop control are complementary, not wholly missing

Mehri, Mehrpouyan, and Chen develop an LSTM/DenseNet-based framework and the Fast LiveStream Predictor for forecasting traffic and congested regions in multi-channel slotted ALOHA. That paper is a prediction contribution, so prediction accuracy must not be treated as evidence of improved deadline-compliant delivery from a controller that was not evaluated there. However, the broader prediction-to-control concept already appears in Jiang et al.'s two-step traffic-prediction/access-control work and the decoupled-learning study. A defensible question is whether prediction is worth its additional complexity for the proposed compact controller, not whether prediction has never been connected to access control.[16][17]

5.5 AMI-specific workload and scheduling studies already exist

Abbas et al. explicitly study NB-IoT smart-meter networks and exploit periodicity to rearrange transmission times and improve uplink-resource utilisation. Stusek et al. evaluate smart-metering performance using NB-IoT control/user-plane optimisation and early data transmission; their stated scenario includes up to 1,000 meters and a 15-minute communication window. These papers invalidate "no work models AMI traffic" as a literature gap. They instead motivate comparing reactive RACH adaptation with application-aware scheduling and making the scale and release assumptions explicit. A result for 1,000 meters under one configuration cannot simply validate 50,000 devices under another.[18][19]

Al-Sammak et al.'s 2025 smart-meter experiments adapt transmission timing using LoRaWAN and NB-IoT hardware. This is adjacent application-side evidence rather than a direct joint-RACH controller baseline. It reinforces the need to distinguish reducing generated traffic from increasing access service capacity.[20]

5.6 Protocol fidelity, coverage, and measurements constrain conclusions

Martiradonna, Piro, and Boggia cross-validate analytical and simulation models of the NB-IoT random-access procedure and explicitly address limitations of simplified or non-standard-compliant models. Liu et al. analyse three coverage-enhancement groups. Agiwal et al. distinguish collision-related failures from poor-channel failures when considering repetitions and power consumption. These studies motivate using a singleton-collision model as an initial abstraction, not as proof of end-to-end standards-compliant performance.[21][22][23]

Caso et al.'s operational-network measurements show that RA outcomes depend on coverage, deployment, and operator configurations, and use ML to support configuration improvement. This is a stronger basis for realistic sensitivity analysis than assuming all failures are collisions or dismissing all prior work as simulation-only. It does not, by itself, validate the proposed controller on a live network.[24]

A particularly relevant recent comparator is Alcaraz et al., *Random Access Control in NB-IoT With Model-Based Reinforcement Learning*, IEEE IoT Journal 12(7), 8135–8149, 2025; DOI 10.1109/JIOT.2024.3499854. Its publisher abstract describes joint NPRACH-parameter and CE-power-threshold adaptation and a model-based method intended to improve early learning efficiency. Therefore, practical joint control and sample efficiency are also active prior-art directions. This assessment is limited to the retrieved abstract/introduction; detailed claims about its reward, deadline accounting, or computational measurements require its full text.[25]

5.7 Synthesis

The reviewed evidence supports neither "all existing methods are static" nor "joint learning-based control is absent." It supports a narrower motivation: evaluate whether an explicitly resource-constrained, compact controller can improve **request-level, deadline-compliant service** for a carefully defined mixture of periodic and event-triggered AMI traffic, under transparent queue accounting, observation limits, and signalling timing. This must be established through matched baselines and ablations, rather than inferred from the use of three action variables.

6. Evidence-based comparison of the closest work

“Not established” below means the examined evidence does not establish the feature; it does not mean the full paper necessarily lacks it.

Work	Established contribution in examined source	Consequence for the revision
Duan et al., D-ACB, 2016	Analytical adaptive barring; fixed and dynamic preamble-allocation variants	Include a strong load-aware analytical baseline; do not call all heuristics static.
Tello-Oquendo et al., ICC 2018	RL adaptation of ACB for LTE-A mixed H2H/M2M traffic	Tabular/RL barring is prior art.
Jiang et al., arXiv:1812.09026	Tabular/linear/DQN control; multi-group, multi-parameter NB-IoT configurations	Closest resource-control comparator; exact action tuple differs from this proposal.
Jiang et al., arXiv:2005.01092	ACB, BO, DQ, multi-objective control, and decoupled prediction/learning	Backoff and coordinated access learning are not absent.
Chen, 2020 thesis	Base-station DRL ACB/backoff control under burst activation and heterogeneous QoS	Include in novelty analysis; a thesis is not an independent journal-paper result.
Bui and Pham, 2020	Dueling DQN jointly controls ACB factor and mean barring time	Distinguish barring time from retry BI.
Hadjadj-Aoul and Ait-Chellouche, 2020	TD3-based NB-IoT blocking control under uncertain access load	Relevant ACB-only neural comparator; deployment cost must be measured, not assumed.
Mehri et al., 2025	Online burst/congestion prediction using LSTM/DenseNet and FLSP	Use only if adding prediction; prediction metrics are not delivery metrics.
Abbas et al., 2020; Stusek et al., 2023	Smart-meter-specific scheduling and NB-IoT capacity/performance studies	AMI modelling and periodicity-aware scheduling already exist.
Martiradonna et al., 2019; Caso et al., 2021 repository version	Protocol-model validation and operational RA measurements	Use to justify model validation and coverage/configuration sensitivity.

Work	Established contribution in examined source	Consequence for the revision
Alcaraz et al., 2025	Publisher abstract establishes model-based joint NPRACH/CE-threshold adaptation	Must be considered before claiming deployable joint control or efficient learning is new.

Source links and evidential limits are given in Section 5. Do not add unverified performance percentages from the existing DOCX reviews to this comparison.

7. Revised research gaps and testable hypotheses

Gap 1 — Deadline-compliant service can be obscured by attempt-level metrics

Bounded gap: The current manuscript and simulator do not establish the proportion of generated AMI requests served on time. Attempt-conditioned success and active-slot delay can conceal deferred demand. The proposed contribution is a transparent request-level evaluation and controller objective—not a claim that deadline-aware networking has never been studied.

Hypothesis H1: Explicit request retention and deadline-aware reward design improve on-time completion relative to collision-oriented and throughput-only rewards, particularly under concentrated arrivals.

Evidence required: request-conservation invariant, per-request timestamps, mature-cohort miss accounting, and reward ablations. If changing the accounting removes the reported benefit, report that result rather than preserving the old numbers.

Gap 2 — The value of this exact joint action set is unproven

Bounded gap: Prior work already combines access-control mechanisms and optimises multiple NB-IoT parameters. The remaining question is whether the **specific feasible combination** of admission, retry timing, and NPRACH configuration provides measurable benefit for this AMI workload at fixed resource cost.

Hypothesis H2: Coordinated three-control adaptation improves the on-time-service/resource trade-off beyond tuned single- and two-control variants.

Evidence required: ACB-only; ACB+backoff; ACB+NPRACH; and full joint-control ablations, with fixed alternatives tuned rather than deliberately weak. Include a source-faithful analytical comparator.

Gap 3 — Compact control under realistic observations and update timing

Bounded gap: True backlog, collision multiplicity, and current demand are not automatically known to an eNB. A small aggregate state may also omit request ages and retry timers. The current paper does not establish that its state is Markov or that its broadcasts can take effect every simulated slot.

Hypothesis H3: A compact controller using implementable observations retains most of its ideal-observation performance when configuration changes are delayed and rate-limited.

Evidence required: observable-versus-oracle state comparison; estimation noise; observation delay; action holding periods; and a documented signalling timeline. If request-age summaries are unavailable without new signalling, mark them as oracle features or implement and cost an estimator.

Gap 4 — AMI workload generalisation, not absence of AMI models

Bounded gap: AMI scheduling has prior art. What remains unestablished here is robustness across a declared combination of reporting periodicity, synchronisation, event size, event recurrence, and coverage conditions.

Hypothesis H4: A policy trained on a defined workload family generalises to held-out event timings and synchronisation levels better than tuned static control, without unacceptable tail-delay degradation.

Evidence required: timestamped per-meter reporting with phase/jitter; event-triggered arrivals superposed on reporting when appropriate; separate training/test traffic seeds; and held-out burst widths, affected fractions, and recurrence patterns. Beta arrivals are an event-timing model, not inherently “non-AMI.” Explain why the selected parameters represent the application.

Gap 5 — Credible performance–cost comparison

Bounded gap: The submitted proxy comparisons do not establish whether tabular control offers a better practical trade-off than actual neural or model-based controllers.

Hypothesis H5: At a stated acceptable loss in on-time completion, compact control reduces measured policy memory and decision latency relative to an actually implemented learning baseline.

Evidence required: CPU model, inference/learning separation, memory including policy overhead, latency distribution, training samples/time, and confidence intervals across independent runs. Do not claim that all DRL requires GPU inference.

Research-gap paragraph suitable for the manuscript

> Prior studies have established adaptive barring, backoff control, multi-parameter NB-IoT resource configuration, prediction-assisted access, and AMI-aware scheduling. Consequently, novelty cannot be inferred from combining reinforcement learning with multiple RACH parameters alone. This study investigates a narrower question: the performance–cost trade-off of a compact network-side policy for deadline-compliant AMI request completion when deferred and retransmitted requests are explicitly retained, resource expenditure is controlled, and observation and configuration-update constraints are represented. The incremental benefit of the selected joint controls and deadline-aware objective will be tested through matched analytical and learning baselines, component ablations, and held-out traffic scenarios. Claims of standards compliance, deadline guarantees, and deployment readiness are conditional on the corresponding protocol mapping and validation evidence.

Novelty status: these are candidate contributions and research hypotheses. The targeted repository review is insufficient to certify an exhaustive “first-ever” claim.

8. Recommended updated solution design

Keep tabular Q-learning as the initial controller, but repair the environment before extending the algorithm.

1. **Arrival generator:** periodic report timestamps with configurable phase/jitter, plus separately parameterised event arrivals. Do not use periodic background

arrivals as a substitute for missing barred requests.

2. **Request-state engine:** explicit barring, retry timers, access stages, completion, deadline expiry, and drops. Check conservation at every step.
3. **Observation interface:** separate simulator truth from measurements/estimates available to the controller.
4. **Action scheduler:** validate legal configurations and model when a new action is signalled and becomes effective. Hold slow controls fixed between allowed updates.
5. **Reward:** start with a documented throughput-only baseline, then test a candidate combining on-time completions, newly missed deadlines, pending-age/backlog cost, and resource cost. Define each weight and normalisation. Such a reward remains a soft optimisation design unless an independent safety/feasibility mechanism is proved.
6. **Learning:** take action, simulate the specified interval, log realised outcomes, compute reward, observe the actual next state, and perform the Q update. Use frozen-policy evaluation separately from exploratory training.
7. **Output:** report request outcomes and policy cost with uncertainty. A deadline-aware state may require more bins/features; recompute table size instead of retaining the old 72 KB claim by assumption.

Recommended working title: *Deadline-Aware Joint Random-Access Control for Dense NB-IoT Smart-Meter Networks: A Lightweight and Reproducible Evaluation.*

Avoid “deployment-ready” and “guaranteed” in the title or abstract at this stage.

9. Minimum validation package before resubmission

9.1 One consistent baseline set

- Tuned fixed configuration—not a supposed universal “3GPP Static” performance level.
- A source-faithful load-aware analytical method, adapted to the same NB-IoT action constraints with differences disclosed.
- ACB-only tabular RL.

- ACB+backoff tabular RL.
- ACB+NPRACH tabular RL.
- Full joint tabular controller.
- At least one genuinely trained DQN or other appropriate learning baseline **if** the paper compares against DRL. Consider model-based RL if its full methods and reproducible implementation can be obtained.

Use the same traffic realisations, resource budgets, observation access, test horizons, and terminal-outcome accounting. Tune methods fairly; equal episode counts alone do not guarantee equal optimisation effort. Report training cost and tuning budget separately.

9.2 Core experiments

Experiment	Question	Main outcomes
Conservation and no-arrival drain tests	Can requests disappear or persist incorrectly?	Exact accounting, final backlog, explicit drops
Simple analytic cross-check	Does singleton contention agree with its known expectation?	Realised attempts and successes versus analytical expectation
Offered-load sweep	Where does demand become infeasible?	On-time completion, backlog growth/drain, unmet demand
Action ablation	Does each control add value?	On-time completion at fixed resource cost
Reward ablation	Is improvement due to useful service or a denominator artefact?	Completion, deadline misses, p95/p99 delay, retained backlog
Update-delay/observability test	Does idealised information or instantaneous control create the gain?	Degradation versus oracle/instantaneous control
Held-out AMI traffic	Does the policy generalise beyond one burst profile?	Per-class on-time completion and tail delay
Coverage/resource sensitivity	Is a collision-only abstraction hiding other bottlenecks?	CE-stratified outcomes and stated protocol limitations

Experiment	Question	Main outcomes
Controller benchmark	Is “lightweight” actually useful?	Policy bytes, CPU decision latency, learning latency, training cost

Use independent training runs and independent held-out tests. Report confidence intervals across independent runs and distinguish seed-to-seed variability from within-run episodes. Report zero misses as an observed sample outcome with cohort size and uncertainty, not proof of zero miss probability.

9.3 Publication integrity checklist

- One registry drives baseline names in prose, tables, and plots.
- No synthetic/proxy learning curves appear as algorithm evidence.
- No invented random-seed spread is used as uncertainty.
- Every metric has an event, denominator, unit, and aggregation rule.
- “Energy” is physical energy only if a supported power/time model exists; otherwise use “attempt proxy.”
- Mean completed-request delay is accompanied by failed/pending requests and tail delay.
- Missing Algorithm 1, disconnected reward arrow, broken references, placeholders, and cropped figures are corrected in the final exported PDF.
- Explain the indicator function correctly and use a coherent SLA definition.
- Release-specific action/signalling mapping supports any standards-compliance statement.
- Reproducible logs, source revision, configurations, and raw results support each numerical claim.

10. Corrections to the repository’s existing reviews

The two DOCX documents should be treated as working notes, not verified secondary sources.

Existing claim or issue	Recommended correction
“No paper models AMI-specific traffic”	Replace with the narrower generalisation question in Gap 4; Abbas and Stusek directly study smart-meter scenarios.
“BI/backoff has been universally ignored”	Remove. Jiang’s decoupled-learning work and Chen’s thesis explicitly address backoff. Distinguish generic BO, barring time, and a standards-specific BI.
“All ML solutions are base-station/cloud-side” and embedded novelty	Not a useful gap for an eNB-side proposal. The repository also contains device-side multi-agent grant-free work, albeit in a different massive-MIMO setting.
“IEC 62056 fixes 15-minute reporting” or “80% within two seconds is a universal hard SLA”	Do not retain without the exact applicable utility specification or standards clause. Communication framework, reporting policy, and operational deadline are different things.
Proposed method is already marked as satisfying hard SLA, embedded deployment, and real-data columns	Replace checkmarks with “proposed” or “not validated.” An AMI simulation is not real measurement data.
Method/reference mismatch in the first review table	The table associates some methods with the wrong reference numbers. Rebuild it from source records rather than editing checkmarks.
Unsupported exact improvements and hardware claims	Recheck against the primary source, scenario, metric, and denominator. Do not carry percentages into the new review merely because they appear in the DOCX.

These corrections arise from comparing the DOCX claims with the primary papers discussed above. Device-side massive-MIMO multi-agent learning is illustrated by Bueno et al.; it is adjacent prior art, not a directly equivalent NB-IoT baseline.[26][27][28]

11. Source management and remaining inputs

Repository organisation for the revised bibliography

- **Primary control literature:** D-ACB; ICC18; ADHOCNets-19; Jiang's 1812.09026, 2002.07759, and 2005.01092 files; Bui/Pham; TD3; Abera et al.; and Chen's thesis chapter on access control.
- **Prediction/measurement/model support:** Mehri et al.; Caso et al.; Martiradonna et al.; three-CE analysis; PHY/MAC overview; reactive-barring inference; power-efficient RAP.
- **AMI context and alternative workload shaping:** Abbas et al.; Stusek et al.; Al-Sammak et al.
- **Background or adjacent—not primary proof of this gap:** generic Q-learning and DQN foundations, broad smart-grid AI and green-IoT architecture reviews, grant-free massive-MIMO methods, and dense heterogeneous-network RA work.
- **Preprint status:** entries such as `2510.25145v1.pdf` identify themselves as submitted/preprint work; do not label them peer-reviewed without verifying publication status.
- **Duplicate control:** `1.pdf` duplicates the long-named Caso PDF; `3.pdf` duplicates `1812.09026v1.pdf`; `2.pdf` duplicates the long-named reactive-barring PDF; `5.pdf` duplicates the long-named multi-channel RL PDF. The Bui, survey, and Wiley/chapter files also include alternate versions or repeated titles. Do not count versions as independent supporting studies.

Specific inputs needed to finalise the research specification

1. **Operational requirement:** which message classes, deadlines, and acceptable miss rates apply? Is 30 seconds a study choice or a utility requirement? Do not substitute the earlier DOCX's two-second claim without a source.
2. **Traffic evidence:** actual/report-configured reporting intervals, event participation fractions, burst widths, recurrence, and whether reporting remains active during an event.
3. **Network configuration:** NB-IoT release, carriers, CE groups, NPRACH opportunities, retry limits, and permitted configuration update mechanisms/timing.

4. **Submission reproducibility:** exact simulator revision, raw logs, learned Q-tables, and figure-generation files used for the rejected manuscript. The present repository does not establish their identity with the submission.
5. **Closest-prior-art access:** full Alcaraz et al. methods and any reproducible implementation, plus verification of the actual LE-URC source.

The revised problem statement and literature synthesis above are usable now. Numerical performance claims, final novelty wording, hard-SLA language, and deployment claims should remain provisional until these inputs and the validation package are resolved.